

The empirical recurrence for OEIS A295210: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A295210 records “Empirical recurrence of order 78”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 98$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 78, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 78 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A295210 is “Number of $n \times 5$ 0..1 arrays with each 1 horizontally or vertically adjacent to 2 or 3 1s.”. It has offset 1 and begins

1, 14, 102, 916, 9536, 85215, 775547, 7213161, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Nov 17 2017 07:23:48, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 98$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 98$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 196$ exact terms of the model returns order 78 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	16	c_{21}	4835372	c_{41}	2816485871	c_{61}	−6750658
c_2	−97	c_{22}	−156076376	c_{42}	−1390696889	c_{62}	9517280
c_3	424	c_{23}	−8798766	c_{43}	−7729317554	c_{63}	−2219125
c_4	−1295	c_{24}	420839135	c_{44}	−1151254828	c_{64}	−3159547
c_5	4118	c_{25}	261657353	c_{45}	−2735312018	c_{65}	861271
c_6	−11960	c_{26}	−425625660	c_{46}	−548777877	c_{66}	663974
c_7	20211	c_{27}	−6110687	c_{47}	668340202	c_{67}	54652
c_8	22051	c_{28}	−21236514	c_{48}	48727851	c_{68}	12607
c_9	−212780	c_{29}	−211458025	c_{49}	2014466966	c_{69}	−11294
c_{10}	221285	c_{30}	−1706360158	c_{50}	309634359	c_{70}	−15997
c_{11}	294699	c_{31}	−939184456	c_{51}	662112147	c_{71}	−6053
c_{12}	66406	c_{32}	−717035863	c_{52}	277222839	c_{72}	−871
c_{13}	−1412105	c_{33}	−1602175156	c_{53}	−30712176	c_{73}	156
c_{14}	−282138	c_{34}	5927220215	c_{54}	384903638	c_{74}	78
c_{15}	7102978	c_{35}	−2031191004	c_{55}	−23876947	c_{75}	16
c_{16}	−674826	c_{36}	9032375401	c_{56}	−67317740	c_{76}	14
c_{17}	−15728393	c_{37}	8340344102	c_{57}	−25463023	c_{77}	3
c_{18}	−4864867	c_{38}	1582723386	c_{58}	−12039855	c_{78}	1
c_{19}	4659376	c_{39}	13794350311	c_{59}	15867034		
c_{20}	24305398	c_{40}	−2543581924	c_{60}	−24676057		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 78, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $78 + 4$ terms removed returns order 78 as well, so $\deg q_{\min} = 78 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $78 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 78 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A295210.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.